Stochastic Calculus: Concise Summary

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Preface

This document provides a highly condensed summary of the main concepts in Stochastic Calculus, suitable for quick revision.

1 Probability Space

Definition: **Sample space**
$\Omega = \{\omega_i : i \in \mathcal{I}\}$ is the set of all possible outcomes of a random experiment.
Definition: **Event**
A subset of Ω.
Definition: **Random variable**
$X : \Omega \to \mathbb{R}$ is a function mapping the sample space to real numbers.
Definition: **Probability measure**
$\mathbb{P}$ on Ω is defined by:

- P1: $\mathbb{P}(\Omega) = 1$
- P2: $0 \leq \mathbb{P}(A) \leq 1$ for any event A
- P3: For disjoint A_i, $\mathbb{P}\left(\bigcup_{i\geq 1} A_i\right) = \sum_{i\geq 1} \mathbb{P}(A_i)$

Theorem: **Probability measure on finite Ω**
If Ω is finite, the probability measure axioms are equivalent to:

- P1*: $\mathbb{P}(\omega_i) \geq 0$ for all i

- P2*: For any event A, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$
- P3*: $\sum_{i=1}^{n} \mathbb{P}(\omega_i) = 1$

Theorem: **Properties of probability measures**

- P4: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- P5: $\mathbb{P}(\varnothing) = 0$
- P6: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- P7: If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$

Definition: **Distribution(law) of a random variable**
The distribution (law) of a random variable X is given by its cumulative distribution function $F_X : \mathbb{R} \to [0, 1]$,
$F_X(x) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \leq x\})$.
Definition: **Distribution(law) for finite Ω**
If Ω is finite, the distribution of X is also given by its probability mass function $f_X : \mathbb{R} \to [0, 1]$,
$f_X(x) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) = x\})$.

Definition: **Equality and equality in distribution** Random variables X and Y are equal if $X(\omega) = Y(\omega)$ for all $\omega \in \Omega$. They are equal in distribution if they have the same distribution.

- If two random variables are equal, then they are equal in distribution.
- Two random variables may be equal in distribution but not equal.
- Two random variables may be equal in distribution under one probability measure but not under another.

Definition: **Sigma-algebra** A σ-algebra $\mathcal{F}$ of Ω is a subset of events such that:

T1: $\Omega \in \mathcal{F}$
T2: If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$
T3: If $A_1, A_2, \ldots \in \mathcal{F}$ then $\bigcup_{i \geq 1} A_i \in \mathcal{F}$

Definition: **Measurable space** The pair $(\Omega, \mathcal{F})$ of a sample space and a σ-algebra is called a measurable space.

Definition: **Finite partition** A family $\mathcal{P} = \{A_1, \ldots, A_n\}$ of events in Ω is a finite partition if:
PF1: $A_i \neq \varnothing$ for all i
PF2: $A_i \cap A_j = \varnothing$ for $i \neq j$
PF3: $\bigcup_{i=1}^{n} A_i = \Omega$

Definition: **Sigma-algebra generated by a partition** : σ-algebra $\mathcal{F}$ is generated from a finite partition $\mathcal{P}$ if it is the smallest σ-algebra containing all elements of $\mathcal{P}$. $\mathcal{F}$ is denoted $\sigma(\mathcal{P})$, and the elements of $\mathcal{P}$ are the atoms of $\mathcal{F}$.

Definition: **Sigma-algebra generated by a random variable** The smallest σ-algebra $\mathcal{F}$ that makes X a random variable on $(\Omega, \mathcal{F})$ is called the σ-algebra generated by X, denoted $\sigma(X)$. If $\mathrm{Card}(\Omega) < \infty$, X takes values $x_1, \ldots, x_m$ and $A_i = \{\omega \in \Omega \mid X(\omega) = x_i\}$, then $\mathcal{P} = \{A_1, \ldots, A_m\}$ is a finite partition and $\sigma(X) = \sigma(\mathcal{P})$.

Definition: **Random variable** A random variable X constructed on the measurable space $(\Omega, \mathcal{F})$ is a real-valued function (maps the sample space to real numbers) $X : \Omega \to \mathbb{R}$ such that for all $x \in \mathbb{R} : \{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$.
if $card(\Omega) < \infty$, then the expression can also be written as $\{\omega \in \Omega : X(\omega) = x\} \in \mathcal{F}$. meaning that X is a random variable on the measurable space $(\Omega, \mathcal{F})$ if and only if the events that characterize its distribution are elements of $\mathcal{F}$ (X **to be $\mathcal{F}$-measurable**).

Theorem: **measurability:** Let $(\Omega, \mathcal{F})$ be a measurable space with $\text{Card}(\Omega) < \infty$ and $\mathcal{P} = \{A_1, \ldots, A_n\}$ the finite partition generating $\mathcal{F}$. The function $X : \Omega \to \mathbb{R}$ is a random variable on that space (X is $\mathcal{F}$-measurable) if and only if X is constant on the atoms of $\mathcal{F}$.

Theorem: **Linear combinations of measurable random variables** If $\text{Card}(\Omega) < \infty$ and $X, Y : \Omega \to \mathbb{R}$ are $\mathcal{F}$-measurable, then for all $a, b \in \mathbb{R}$, $aX + bY$ is also $\mathcal{F}$-measurable. Any linear combination of random variables on the same measurable space is a random variable of that space.

Definition: **Probability measure** Let $(\Omega, \mathcal{F})$ be a measurable space. A function $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure if:
P1: $\mathbb{P}(\Omega) = 1$
P2: $0 \leq \mathbb{P}(A) \leq 1$ for all $A \in \mathcal{F}$
P3: For disjoint $A_i \in \mathcal{F}$, $\mathbb{P}(\bigcup_{i \geq 1} A_i) = \sum_{i \geq 1} \mathbb{P}(A_i)$

Definition: **Probability space** The triple $(\Omega, \mathcal{F}, \mathbb{P})$ of a sample space, a σ-algebra, and a probability measure is called a probability space.

2 Stochastic Processes

Definition: **Stochastic process** Let $(\Omega, \mathcal{F})$ be a measurable space. A stochastic process $X = \{X_t : t \in \mathcal{T}\}$ is a family of random variables, all built on the same measurable space $(\Omega, \mathcal{F})$, where $\mathcal{T}$ is a set of indices.

Definition: **Filtration** A family $\mathbb{F} = \{\mathcal{F}_t : t \in \mathcal{T}\}$ of σ-algebras on Ω is a filtration if:

F1: $\mathcal{F}_t \subseteq \mathcal{F}$ for all t
F2: If $t_1 \leq t_2$, then $\mathcal{F}_{t_1} \subseteq \mathcal{F}_{t_2}$

Definition: **Adapted process** A stochastic process $X = \{X_t : t \in \mathcal{T}\}$ is adapted to the filtration $\mathbb{F} = \{\mathcal{F}_t : t \in \mathcal{T}\}$ if X_t is $\mathcal{F}_t$-measurable for all t.

Definition: **Generated filtration** The filtration $\mathbb{F} = \{\mathcal{F}_t : t \in \mathcal{T}\}$ is generated by the process $X = \{X_t : t \in \mathcal{T}\}$ if $\mathcal{F}_t = \sigma(\{X_s : s \leq t\})$ for all t.

Definition: **Stopping time** Let $(\Omega, \mathcal{F})$ be a measurable space with $\text{Card}(\Omega) < \infty$ and filtration $\mathbb{F} = \{\mathcal{F}_t : t \in \{0, 1, \ldots\}\}$. A stopping time τ is an $(\Omega, \mathcal{F})$-random variable taking values in $\{0, 1, \ldots\}$ such that $\{\omega \in \Omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$ for all t. In discrete time,

it is equivalent to check $\{\tau = t\}$ for each t. Intuitively, the time τ when one makes a decision is a stopping time if the decision is made based on the information available at that time.

Theorem: If τ_1 and τ_2 are stopping times, then $\tau_1 \wedge \tau_2 \equiv \min\{\tau_1, \tau_2\}$ and $\tau_1 \vee \tau_2 \equiv \max\{\tau_1, \tau_2\}$ are also stopping times.

Definition: **First hitting time:** Let $(\Omega, \mathcal{F})$ be a measurable space with $\text{Card}(\Omega) < \infty$ and filtration $\mathbb{F} = \{\mathcal{F}_t : t \in \{0, 1, \ldots\}\}$. Let $X = \{X_t : t \in \{0, 1, \ldots\}\}$ be a stochastic process adapted to $\mathbb{F}$, and $B \subseteq \mathbb{R}$. The first time X enters B is $\tau_B(\omega) = \min\{t \in \{0, 1, \ldots\} : X_t(\omega) \in B\}$.
If $X_t(\omega)$ never hits B, define $\tau_B(\omega) = \infty$.

Theorem: The first hitting time τ_B is a stopping time.

3 Conditional Expectation

Definition: **Independence of events:** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Two events A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Definition: **Independence of random variables:** Random variables X and Y are independent if for all $A \in \mathcal{P}_X$ and $B \in \mathcal{P}_Y$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, where $\mathcal{P}_X$ and $\mathcal{P}_Y$ are finite partitions generating $\sigma(X)$ and $\sigma(Y)$. Intuitively, X and Y are independent if knowing one gives no information about the other.

Definition: **Conditional probability:** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For any event $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, the conditional probability given A is $\mathbb{P}(B|A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}$ for all $B \in \mathcal{F}$.

Theorem: **Independence via conditional probability** Let X and Y be random variables built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, $\text{Card}(\Omega) < \infty$. Let $\mathcal{P}_X = \{A_1, \ldots, A_m\}$ and $\mathcal{P}_Y = \{B_1, \ldots, B_n\}$ be finite partitions generating $\sigma(X)$ and $\sigma(Y)$. If

$$\forall A \in \mathcal{P}_X \text{ such that } \mathbb{P}(A) > 0, \mathbb{P}(B|A) = \mathbb{P}(B), \forall B \in \mathcal{P}_Y$$

or

$$\forall B \in \mathcal{P}_Y \text{ such that } \mathbb{P}(B) > 0, \mathbb{P}(A|B) = \mathbb{P}(A), \forall A \in \mathcal{P}_X$$

then X and Y are independent.

Definition: **Expectation:** Let X be a random variable built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\text{Card}(\Omega) < \infty$. The expectation of X, denoted $\mathbb{E}^{\mathbb{P}}[X]$ is

$$\mathbb{E}^{\mathbb{P}}[X] = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega).$$

Properties: Let X and Y be two random variables built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, $\text{Card}(\Omega) < \infty$. If a and b represent real numbers, then

E1 $\mathbb{E}^{\mathbb{P}}[aX + bY] = a\mathbb{E}^{\mathbb{P}}[X] + b\mathbb{E}^{\mathbb{P}}[Y]$;
E2 If $\forall \omega \in \Omega$, $X(\omega) \leq Y(\omega)$ then $\mathbb{E}^{\mathbb{P}}[X] \leq \mathbb{E}^{\mathbb{P}}[Y]$;
E3 In general, $\mathbb{E}^{\mathbb{P}}[XY] \neq \mathbb{E}^{\mathbb{P}}[X]\mathbb{E}^{\mathbb{P}}[Y]$;
E4 If X and Y are independent, then $\mathbb{E}^{\mathbb{P}}[XY] = \mathbb{E}^{\mathbb{P}}[X]\mathbb{E}^{\mathbb{P}}[Y]$.

Definition: **kth moment** Let X be a random variable built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathrm{Card}(\Omega) < \infty$. The kth moment of X, denoted $\mathbb{E}^{\mathbb{P}}\left[X^k\right]$ is the expectation of the variable X^k:

$$\mathbb{E}^{\mathbb{P}}\left[X^k\right] = \sum_{\omega \in \Omega} X^k(\omega)\mathbb{P}(\omega) = \sum_{i=1}^{n} x_i^k f_X(x_i)$$

where $x_1 < \ldots < x_n$ are the values taken by X and f_X is its probability mass function.

Definition: **Variance** Let X be a random variable built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathrm{Card}(\Omega) < \infty$. The variance of X, denoted $\mathrm{Var}^{\mathbb{P}}[X]$ is the expectation of the random variable $\left(X - \mathbb{E}^{\mathbb{P}}[X]\right)^2$:

$$\mathrm{Var}^{\mathbb{P}}[X] = \sum_{\omega \in \Omega} \left(X(\omega) - \mathbb{E}^{\mathbb{P}}[X]\right)^2 \mathbb{P}(\omega) \tag{1}$$

$$= \sum_{i=1}^{n} \left(x_i - \mathbb{E}^{\mathbb{P}}[X]\right)^2 f_X(x_i) \tag{2}$$

where $x_1 < \ldots < x_n$ are the values taken by X and f_X is its probability mass function.

Properties:
V1 $\mathrm{Var}^{\mathbb{P}}[X] \geq 0$;
V2 $\mathrm{Var}^{\mathbb{P}}[X] = \mathbb{E}^{\mathbb{P}}[X^2] - \left(\mathbb{E}^{\mathbb{P}}[X]\right)^2$;
V3 $\forall a, b \in \mathbb{R}$, $\mathrm{Var}^{\mathbb{P}}[aX + b] = a^2 \mathrm{Var}^{\mathbb{P}}[X]$;
V4 If X and Y are independent, then $\mathrm{Var}^{\mathbb{P}}[X + Y] = \mathrm{Var}^{\mathbb{P}}[X] + \mathrm{Var}^{\mathbb{P}}[Y]$.

Definition: **Covariance** Let X and Y be two random variables built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathrm{Card}(\Omega) < \infty$. The covariance of X and Y, denoted $\mathrm{Cov}^{\mathbb{P}}[X, Y]$, is the expectation of the random variable $(X - \mathbb{E}^{\mathbb{P}}[X])(Y - \mathbb{E}^{\mathbb{P}}[Y])$:

$$\mathrm{Cov}^{\mathbb{P}}[X, Y] = \sum_{\omega \in \Omega} \left(X(\omega) - \mathbb{E}^{\mathbb{P}}[X]\right)\left(Y(\omega) - \mathbb{E}^{\mathbb{P}}[Y]\right) \mathbb{P}(\omega).$$

Properties:

C1 $\mathrm{Cov}^{\mathbb{P}}[X, Y] = \mathbb{E}^{\mathbb{P}}[XY] - \mathbb{E}^{\mathbb{P}}[X]\,\mathbb{E}^{\mathbb{P}}[Y]$;
C2 If X and Y are independent, then $\mathrm{Cov}^{\mathbb{P}}[X, Y] = 0$;
C3 $\forall a, b \in \mathbb{R}$, $\mathrm{Cov}^{\mathbb{P}}[aX_1 + bX_2, Y] = a\,\mathrm{Cov}^{\mathbb{P}}[X_1, Y] + b\,\mathrm{Cov}^{\mathbb{P}}[X_2, Y]$;
C4 $\mathrm{Var}^{\mathbb{P}}[X + Y] = \mathrm{Var}^{\mathbb{P}}[X] + \mathrm{Var}^{\mathbb{P}}[Y] + 2\,\mathrm{Cov}^{\mathbb{P}}[X, Y]$.

Definition: **Conditional expectation:** Let the random variable X be built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be the σ-algebra generated by the finite partition

$\mathcal{P} = \{A_1, \ldots, A_n\}$ satisfying $\forall i \in \{1, \ldots, n\}$, $\mathbb{P}(A_i) > 0$. The conditional expectation of X given $\mathcal{G}$, denoted $\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}]$, is

$$\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}](\omega) = \sum_{i=1}^{n} \frac{\mathbb{I}_{A_i}(\omega)}{\mathbb{P}(A_i)} \sum_{\omega^* \in A_i} X(\omega^*)\, \mathbb{P}(\omega^*) = \sum_{i=1}^{n} \mathbb{I}_{A_i}(\omega) \sum_{\omega^* \in A_i} X(\omega^*)\, \mathbb{P}(\omega^* | A_i).$$

where $\mathbb{I}_{A_i} : \Omega \to \{0, 1\}$ is the indicator function

$$\mathbb{I}_{A_i}(\omega) = \begin{cases} 1 & \text{if } \omega \in A_i, \\ 0 & \text{if } \omega \notin A_i. \end{cases}$$

Properties: Let $\mathcal{G}$ be the σ-algebra generated by the finite partition $\mathcal{P} = \{A_1, \ldots, A_n\}$.

Then for any random variables X, Y built on $(\Omega, \mathcal{F}, \mathbb{P})$:
EC1 If X is $\mathcal{G}$-measurable, then $\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}] = X$;
EC2 If $\mathcal{G}_1 \subseteq \mathcal{G}_2$ are sigma-algebras, then $\mathbb{E}^{\mathbb{P}}\big[\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}_1] \mid \mathcal{G}_2\big] = \mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}_1]$;
EC3 If $\mathcal{G}_1 \subseteq \mathcal{G}_2$ are sigma-algebras, then $\mathbb{E}^{\mathbb{P}}\big[\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1\big] = \mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}_1]$;
EC4 $\mathbb{E}^{\mathbb{P}}[X \mid \{\varnothing, \Omega\}] = \mathbb{E}^{\mathbb{P}}[X]$;
EC5 $\mathbb{E}^{\mathbb{P}}\big[\mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}]\big] = \mathbb{E}^{\mathbb{P}}[X]$;
EC6 If Y is $\mathcal{G}$-measurable, then $\mathbb{E}^{\mathbb{P}}[XY \mid \mathcal{G}] = Y\, \mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}]$;
EC7 If X and Y are independent, then $\mathbb{E}^{\mathbb{P}}[X \mid \sigma(Y)] = \mathbb{E}^{\mathbb{P}}[X]$;
EC8 $\forall a, b \in \mathbb{R}$, $\mathbb{E}^{\mathbb{P}}[aX + bY \mid \mathcal{G}] = a\, \mathbb{E}^{\mathbb{P}}[X \mid \mathcal{G}] + b\, \mathbb{E}^{\mathbb{P}}[Y \mid \mathcal{G}]$.

4 Martingale

Definition: **Discrete-time martingale**
On the filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{F}$ is the filtration $\{\mathcal{F}_t : t \in \{0, 1, 2, \ldots\}\}$, the stochastic process

$$M = \{M_t : t \in \{0, 1, 2, \ldots\}\}$$

is a discrete-time *martingale* if:

- (M1) $\forall t \in \{0, 1, 2, \ldots\}$, $\mathbb{E}^{\mathbb{P}}[|M_t|] < \infty$ (being integrable)
- (M2) $\forall t \in \{0, 1, 2, \ldots\}$, M_t is $\mathcal{F}_t$–measurable
- (M3) $\forall s, t \in \{0, 1, 2, \ldots\}$ such that $s < t$, $\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_s] = M_s$

Lemma: **Equivalent third condition**
In the definition of a martingale, the condition (M3) is equivalent to

$$\text{M3*} \quad \forall t \in \{1, 2, \ldots\}, \quad \mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1}.$$

Definition: **Stopped process**
The stochastic process X and the stopping time τ are built on the same filtered measurable space $(\Omega, \mathcal{F}, \mathbb{F})$. The stochastic process X^τ defined by

$$X_t^\tau(\omega) = X_{t \wedge \tau(\omega)}(\omega)$$

is called a *stopped process* with stopping time τ.

Theorem: **Stopped martingale**

If the martingale M and the stopping time τ are built on the same filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, then the stopped process M^τ is also a martingale on that space.

Theorem: **Optional Stopping Theorem**

Let

$$X = \{X_t : t \in \{0, 1, 2, \ldots\}\}$$

be a process built on the filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{F}$ is the filtration $\{\mathcal{F}_t : t \in \{0, 1, 2, \ldots\}\}$. Let's assume that the stochastic process X is $\mathbb{F}$–adapted and that it is integrable, i.e. $\mathbb{E}^{\mathbb{P}}[|X_t|] < \infty$. Then X is a martingale if and only if

$$\mathbb{E}^{\mathbb{P}}[X_\tau] = \mathbb{E}^{\mathbb{P}}[X_0]$$

for any bounded stopping time τ, i.e. for any given stopping time τ, there exists a constant b such that

$$\forall \omega \in \Omega, \quad 0 \leq \tau(\omega) < b.$$

Definition: **Markovian process**

A stochastic process $X = \{X_t : t \in \mathcal{T}\}$, where $\mathcal{T}$ is a set of indices, is said to be *markovian* if, for any $t_1 < t_2 < \ldots < t_n \in \mathcal{T}$, the conditional distribution of X_{t_n} given $X_{t_1}, \ldots, X_{t_{n-1}}$ is equal to the conditional distribution of X_{t_n} given $X_{t_{n-1}}$, i.e. for any $x_1, \ldots, x_n \in \mathbb{R}$,

$$\mathbb{P}[X_{t_n} \leq x_n \mid X_{t_1} = x_1, \ldots, X_{t_{n-1}} = x_{n-1}] = \mathbb{P}[X_{t_n} \leq x_n \mid X_{t_{n-1}} = x_{n-1}].$$

Definition: **Random walk**

The stochastic process X, built on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, is a *random walk* if it admits the representation

$$X_0 = 0 \quad \text{and} \quad \forall t \in \{1, 2, \ldots\}, \quad X_t = \sum_{n=1}^{t} \xi_n$$

where the sequence $\{\xi_i : i \in \{1, 2, \ldots\}\}$ is made up of independent and identically distributed random variables. Random walks are Markovian processes.

5 Discrete Time Market Model

Riskless: $S_t^0 = (1+r)^t$, $S_0^0 = 1$. Risky: S_t^1 binary $\{s_1, s_2\}$, initial s_0. Portfolio $\phi = (\phi_0, \phi_1)$: $V_t(\phi) = \phi_0 S_t^0 + \phi_1 S_t^1$, cost $V_0(\phi) = \phi_0 + \phi_1 s_0$.

Definition: **Arbitrage opportunity** A portfolio ϕ is an arbitrage opportunity if:

A1: $V_0(\phi) = 0$ (zero initial investment)
A2: $\forall \omega \in \Omega$, $V_1(\omega) \geq 0$ (zero probability of losing money)
A3: $\exists \omega \in \Omega$, $V_1(\omega) > 0$ (positive probability of gaining)

So, arbitrage is possible if and only if $s_2 > s_1 \geq s_0(1+r)$ or $s_1 < s_2 \leq s_0(1+r)$. We assume the model contains no arbitrage opportunities, i.e., $s_1 < s_0(1+r) < s_2$.

Definition: **Contingent claim:** A contingent claim is a contract (seller and buyer) whose value depends on the market state during the contract validity period (like insurance). Mathematically, a contingent claim C is any non-negative random variable.

Replicating portfolio: $\phi = (\phi_0, \phi_1)$ replicates contingent claim C if $V_1(\phi) = C$. If replicating portfolio exists, $\phi_0 = \frac{s_0 c_1 - s_1 c_2}{(s_2 - s_1)(1+r)}$ and $\phi_1 = \frac{c_2 - c_1}{s_2 - s_1}$.

Risk-neutral measure Probability measure $\mathbb{Q}$ defined by $q = \mathbb{Q}(\omega_1) = \frac{s_2 - s_0(1+r)}{s_2 - s_1}$ and $1 - q = \mathbb{Q}(\omega_2)$. Under no-arbitrage, $0 < q < 1$.

Martingale property Under $\mathbb{Q}$: expected return on risky security $= r$ (same as riskless). Discounted price process $\frac{S_t^1}{(1+r)^t}$ is $\mathbb{Q}$-martingale.

Definition: **Return on a security:** The return on a security from $t-1$ to t

$$R_t(\omega) = \frac{S_t(\omega) - S_{t-1}(\omega)}{S_{t-1}(\omega)}$$

Definition: **Price process** The $(K+1)$-dimensional stochastic process $\mathbf{S} = \{\mathbf{S}_t\}_{t \in \{0,1,\ldots,T\}}$ representing the evolution of individual unit prices of all securities is made up of random column vectors $\mathbf{S}_t = (S_t^0, S_t^1, \ldots, S_t^K)^\top$, where S_t^k is the unit price of security k at time t.

Definition: **Riskless security** The 0th security plays a peculiar role in that it is a riskless security (e.g., bank account or bond). It implies that $\forall \omega \in \Omega$ and $\forall t \in \{0, 1, \ldots, T-1\}$: $S_t^0(\omega) \leq S_{t+1}^0(\omega)$. Without loss of generality, $S_0^0(\omega) = 1$.

Definition: **Discount factor** The unidimensional stochastic process $\beta_t = \frac{1}{S_t^0}$ is used as discount factor.

Definition: **Trading strategy:** A trading strategy is a $\{\mathcal{F}_t\}_{t \in \{1,\ldots,T\}}$-predictable vector stochastic process $\phi = \{\phi_t\}_{t \in \{1,\ldots,T\}}$, where the random row vector $\phi_t = (\phi_t^0, \phi_t^1, \ldots, \phi_t^K)$ represents the investor's portfolio at time t. Here, ϕ_t^k is the number of shares of security k held during $(t-1, t]$.

Definition: **Self-financing investment strategy**

A strategy is self-financing if no funds are added to or withdrawn from the value of the portfolio after time $t = 0$, i.e., $\forall t \in \{1, \ldots, T-1\}$: $\phi_t \mathbf{S}_t = \phi_{t+1} \mathbf{S}_t$. assuming zero transaction costs.

Definition: **Market value**

The process $V(\phi) = \{V_t(\phi) : t \in \{0, 1, \ldots, T\}\}$ represents the market value of the strategy at any time:

$$V_t(\phi) = \begin{cases} \phi_1 \mathbf{S}_0 & \text{if } t = 0 \\ \phi_t \mathbf{S}_t & \text{if } t \in \{1, \ldots, T\} \end{cases}$$

Definition: **Admissible investment strategies**

A trading strategy is called admissible if it is self-financing and its market value is never negative. The set Φ of admissible strategies is:

$$\Phi = \left\{ \phi \;\middle|\; \begin{array}{l} \text{each component } \phi^k \text{ of } \phi \text{ is predictable,} \\ \phi \text{ is self-financing,} \\ \text{and } \forall t \in \{0, \ldots, T\} : V_t(\phi) \geq 0 \end{array} \right\}$$

Definition: **Arbitrage opportunity**

An admissible strategy ϕ is an arbitrage opportunity if $V_0(\phi) = 0$ and if $\mathbb{E}^{\mathbb{P}}[V_T(\phi)] > 0$.

Definition: **Contingent claim**

A contingent claim X is a $(\Omega, \mathcal{F}_T)$-non-negative random variable. It can be thought of as a contract between two parties, such that $X(\omega)$ will be paid by one party to the other if state ω pertains. The set $\mathbf{X}$ of contingent claims is:

$$\mathbf{X} = \left\{ X \;\middle|\; \begin{array}{l} X \text{ is a } (\Omega, \mathcal{F}_T)\text{-random variable} \\ \text{such that } \forall \omega \in \Omega, X(\omega) \geq 0 \end{array} \right\}$$

Definition: **Attainable contingent claim**

A contingent claim X is said to be attainable if there exists an admissible trading strategy ϕ that can replicate the cash flow generated by X, i.e., $V_T(\phi) = X$.

Definition: **Price system**

A price system π is a linear operator on $\mathbf{X}$ that returns non-negative values $\pi : \mathbf{X} \to [0, \infty)$ and that satisfies:

$$\pi(X) = 0 \Leftrightarrow X = 0$$

and

$$\forall a, b \geq 0 \text{ and } \forall X_1, X_2 \in \mathbf{X}, \quad \pi(aX_1 + bX_2) = a\pi(X_1) + b\pi(X_2).$$

Intuition: We are implicitly assuming that we are small investors so if we buy more the price won't go up (orderbook effect) or buying more won't make the price less (like Costco effect :))

Definition: **Consistency**

A price system is said to be consistent with the market model if the price associated with the contingent claim $V_T(\phi)$ is its market value at time $t = 0$, $V_0(\phi)$, i.e. $\pi(V_T(\phi)) = V_0(\phi)$. The set Π of price systems consistent with the market model is:

$$\Pi = \left\{ \pi : \mathbf{X} \to [0, \infty) \;\middle|\; \begin{array}{c} \pi(X) = 0 \Leftrightarrow X = 0, \\ \forall a, b \geq 0 \text{ and } \forall X_1, X_2 \in \mathbf{X}, \\ \pi(aX_1 + bX_2) = a\pi(X_1) + b\pi(X_2), \\ \forall \phi \in \Phi,\ \pi(V_T(\phi)) = V_0(\phi) \end{array} \right\}$$

Definition: **Equivalent martingale measures**

Let $\mathbf{P}$ denote the set of martingale measures equivalent to $\mathbb{P}$:

$$\mathbf{P} = \left\{ \mathbb{Q} \,\middle|\, \begin{array}{l} \mathbb{Q} \text{ is a probability measure on } (\Omega, \mathcal{F}), \\ \forall \omega \in \Omega, \mathbb{Q}(\omega) > 0, \text{ and} \\ \forall k \in \{0, 1, \ldots, K\},\ \beta S^k \text{ is a } \mathbb{Q}\text{-martingale} \end{array} \right\}$$

Definition: **Predictable process**

A stochastic process $X = \{X_t : t = 0, 1, 2, \ldots\}$ built on a measurable space $(\Omega, \mathcal{F})$ is said to be predictable with respect to the filtration $\{\mathcal{F}_t : t = 0, 1, 2, \ldots\}$ if X_0 is a $(\Omega, \mathcal{F}_0)$-random variable and $\forall t \in \{1, 2, \ldots\}$, X_t is a $(\Omega, \mathcal{F}_{t-1})$-random variable.

Theorem: **Bijective correspondence (Proposition 2.6)**

There is bijective correspondence between the set Π of price systems that are consistent with the market model and the set $\mathbf{P}$ of martingale measures equivalent to $\mathbb{P}$. Such a correspondence is given by:

(i) $\pi(X) = \mathbb{E}^{\mathbb{Q}}[\beta_T X]$, $X \in \mathbf{X}$

(ii) $\mathbb{Q}(A) = \pi(S_T^0 \mathbb{I}_A)$, $A \in \mathcal{F}$

Intuition: if there exists a probability measure under which all discounted price processes are martingales, then we can build, based on such a measure, a price system for attainable contingent claims that is consistent with the market model. and vice versa.

Definition: **Equivalence of two probability measures**

Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures that exist on the measurable space $(\Omega, \mathcal{F})$. The measures $\mathbb{P}$ and $\mathbb{Q}$ are said to be equivalent if and only if the impossible events are the same under both measures, i.e.,

$$\forall A \in \mathcal{F}, \quad \mathbb{P}(A) = 0 \Leftrightarrow \mathbb{Q}(A) = 0.$$

In our case, since $\forall \omega \in \Omega$, $\mathbb{P}(\omega) > 0$, all measures $\mathbb{Q}$ equivalent to $\mathbb{P}$ shall satisfy the condition

$$\forall \omega \in \Omega, \quad \mathbb{Q}(\omega) > 0.$$

Theorem: **Lemma on page 228**

If there exists a self-financing strategy ϕ (not necessarily admissible) such that $V_0(\phi) = 0$, $V_T(\phi) \geq 0$ and $\mathbb{E}^{\mathbb{P}}[V_T(\phi)] > 0$, then there exists an arbitrage opportunity.

Interpretation:

- This lemma provides us with a necessary and sufficient condition for the existence of an arbitrage opportunity.
- The advantage of this result is that, from now on, we don't have to verify any more whether the strategy that is a candidate for an arbitrage opportunity is admissible.

Theorem: **Fundamental theorem of asset pricing (Theorem 2.7)**

The market model contains no arbitrage opportunities if and only if there exists at least one martingale measure equivalent to $\mathbb{P}$.

Corollary: **The law of one price**

If the market model contains no arbitrage, then there is a single price associated with any attainable contingent claim X and it satisfies $\pi = \mathbb{E}^{\mathbb{Q}}[\beta_T X]$ for each martingale measure $\mathbb{Q} \in \mathbf{P}$.

Theorem: **Proposition 2.8**

If ϕ is an admissible strategy, then the process $\beta V(\phi)$, representing its discounted market value, is a $\mathbb{Q}$-martingale for each measure $\mathbb{Q} \in \mathbf{P}$.

Definition: **Complete market**

A market is said to be complete if it contains no arbitrage opportunity and if all contingent claims are attainable.

Theorem: **Characterization of complete markets**

A market is complete if and only if there exists a unique martingale measure.

6 Brownian Motion

7 Stochastic Integral